

Active Perception of Object Shape Using a Sensorless Dexterous Robotic Hand

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Abstract

Robots commonly rely on cameras, depth sensors, or dedicated tactile sensors to identify and understand objects. In this project, we investigate whether a dexterous robotic hand can identify an object's shape and estimate its dimensions using proprioceptive information obtained through physical interaction, without relying on external visual or dedicated tactile sensing.

We use a LEAP Hand to interact with two geometric objects: a cube with a side length of 7.5 cm and a sphere with a diameter of 7.5 cm. The hand performs controlled exploratory movements while we record proprioceptive information such as joint states, motor positions, finger trajectories, and, where applicable, motor currents. These observations provide indirect information about the object's geometry through the way the fingers move and interact with its surface.

We follow an active perception approach in which manipulation is treated as a source of information. Rather than making a decision from a single grasp, the hand explores the object through successive interactions and uses the resulting observations to construct an initial internal representation. As manipulation continues, new contacts generated through movements such as rotation, sliding, pivoting, or regrasping provide additional information that can be used to refine this representation. We then identify the object as a cube or a sphere and estimate its corresponding physical dimension.

Keywords: Active perception, dexterous manipulation, proprioception, robotic hand, object shape recognition, LEAP Hand, proprioceptive perception.

1. Introduction

Understanding an unfamiliar object is an important capability for a robot that needs to interact with its surroundings. Humans can often infer an object’s approximate shape and size through physical interaction by moving their fingers, changing their grasp, and observing how the object responds, even when visual information is unavailable. This raises the possibility that a robotic hand could similarly obtain information about an object through its own movements and interactions.

Dexterous robotic hands provide a suitable platform for investigating this idea. Their multiple fingers and joints generate proprioceptive information about the configuration and movement of the hand. When the fingers interact with an object, these measurements can indirectly reflect characteristics of its geometry.

Previous studies have demonstrated the potential of proprioception for object recognition. V’asquez and Perdereau developed a proprioceptive shape signature based on joint angles to recognize primitive object shapes and estimate object pose. Their representation was designed to be invariant to object size and position (?). Sintov and Meir subsequently demonstrated object recognition using the kinematics of a multi-fingered robotic hand without relying on visual or tactile sensing, including recognition of broader geometric categories (?).

These studies establish that proprioceptive information can provide useful information about object geometry. However, their primary focus is on recognizing object shapes or categories, rather than estimating the physical dimensions of the object as an output. The present project builds on this direction by investigating whether proprioceptive interaction can be used for both shape identification and dimensional estimation.

In the present study, we investigate whether a dexterous robotic hand can identify an object as a cube or sphere and estimate its physical dimension using proprioceptive information obtained through active physical interaction, without external visual or dedicated tactile sensing. We use a LEAP Hand and two geometric objects: a cube with a side length of 7.5 cm and a sphere with a diameter of 7.5 cm. Their known dimensions provide ground truth against which the estimated dimensions can be evaluated.

2. Data and Methods

2.1. *Experimental Setup*

The experiments were conducted using a LEAP Hand, an open-source dexterous robotic hand with four fingers and 16 actuated joints. The index, middle, and ring fingers each have four joints, while the thumb has four joints comprising base rotation and flexion-related movements. The hand provides proprioceptive information about its joint configuration during physical interaction with an object.

Two geometric objects were used in the experiments: a cube with a side length of 7.5 cm and a sphere with a diameter of 7.5 cm. The known dimensions of these objects provide ground-truth values for evaluating the estimated dimensions.

The hand was operated using the Ex-ARM Python control framework developed for the LEAP Hand. The framework provides an interface for hardware control and includes a kinematic model for calculating fingertip positions from the joint configuration. The hardware interface provides access to joint positions, velocities, and motor currents, while the kinematic model provides forward kinematics for the four fingertips.

2.2. *Proprioceptive Data*

The perception process relies on information obtained from the robotic hand itself rather than from external cameras, depth sensors, or dedicated tactile sensors. During physical interaction, the configuration of the fingers changes according to the geometry of the contacted surface. These changes provide indirect information about the shape and dimensions of the object.

The primary proprioceptive measurements consist of joint states obtained from the LEAP Hand. Motor-current measurements are also used to identify changes associated with physical contact. Since motor current is not a dedicated tactile measurement, it is treated as a contact proxy rather than as a direct measurement of contact force.

The recorded joint positions are supplied to the LEAP Hand forward kinematic model to determine the corresponding fingertip positions in the hand coordinate frame. The resulting fingertip locations provide spatial observations of the object surface during interaction.

2.3. Active Physical Exploration

The object is explored through a sequence of physical interactions rather than through a single static grasp. The fingers are moved towards the object and contact is identified from changes in the proprioceptive state. For each detected contact, the corresponding joint configuration is used to determine the fingertip position through forward kinematics.

The hand and object are maintained in a fixed relative configuration during individual probing operations, and the hand is released between successive contacts. This allows contact observations to be collected from different probing configurations while maintaining a common reference frame.

Each new contact provides an additional observation of the object’s surface. Consequently, the geometric representation is progressively constructed from multiple contact locations. Movements that change the contact configuration can expose different portions of the object and provide additional information for perception.

This approach treats manipulation as part of the perception process. Instead of separating perception from manipulation, the physical interaction itself provides the observations from which the object’s geometry can be inferred.

2.4. Object Shape Identification

The collected proprioceptive observations are used to distinguish between the two object geometries. The fingertip contact locations obtained from the hand’s joint states provide a spatial representation of the contacted surface.

For the sphere, contact locations are expected to follow the curved surface of the object, whereas contacts with the cube are associated with its planar faces and edges. The spatial distribution of the contact observations is therefore used to distinguish the two geometric classes.

The Ex-ARM implementation includes an experimental proprioceptive shape-recognition procedure based on LEAP joint states and motor-current changes for distinguishing sphere-like and box-like objects. Contact samples are generated from the measured joint positions using the LEAP Hand forward-kinematics model.

2.5. Dimensional Estimation

Following shape identification, the proprioceptive contact observations are used to estimate the corresponding physical dimension of the object. For

the cube, the characteristic dimension is its side length, while for the sphere it is its diameter.

The estimated dimension is obtained from the spatial information provided by the successive contact observations and is compared with the known ground-truth dimension of 7.5 cm. This provides a direct measure of how well proprioceptive interaction can recover the physical scale of the object without external visual or dedicated tactile sensing.

The complete implementation used in this study is publicly available in the Ex-ARM GitHub repository (?).

3. Results